

UMME AMMARA

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RESEARCH INTERESTS

Human-Computer Interaction; human-centered and responsible AI; AI-mediated healthcare; ICTD; technology and work; AI-assisted decision-making.

EDUCATION

- | | |
|---|---|
| Georgia Institute of Technology
<i>M.S. in Human-Computer Interaction GPA: 4.0</i> <ul style="list-style-type: none">MS Foley Scholar Finalist. | Aug 2024 – May 2026
Atlanta, GA |
| Lahore University of Management Sciences (LUMS)
<i>B.S. in Computer Science GPA: 3.7</i> <ul style="list-style-type: none">Graduated with Distinction; Dean's Honor List (2018–2022). | Aug 2018 – Jun 2022
Lahore, Pakistan |

PUBLICATIONS

- Unpacking Space, Place, and Labor in Experiences of Menopause**
Umme Ammara, Fozia Umer Qureshi, and Naveena Karusala.
ACM CHI 2026. doi:10.1145/3772318.3791284
- Awaaz-e-Sehat: A Mobile Voice-Based AI System for EMR Generation and Clinical Decision Support in Low-Resource Maternal Healthcare**
Maryam Mustafa, Amna Shah Nawaz, Umme Ammara, Moaiz Abrar, Bakhtawar Ahtisham, Fozia Umer Qurashi, Mostafa Shahin, and Beena Ahmed.
ACM IMWUT 2026. doi:10.1145/3790115

RESEARCH EXPERIENCE

- | | |
|---|--|
| Georgia Institute of Technology
<i>Graduate Research Assistant, Technologies & International Development (T+ID) Lab</i> <ul style="list-style-type: none">Contributed to a collaborative research initiative with Emory University and Morehouse School of Medicine to design digital tools supporting targeted and adaptive health messaging through SMS.Developed interview protocols and conducted think-aloud walkthroughs with 29 Master of Public Health trainees and faculty to evaluate a health messaging dashboard and practitioner workflows for targeted messaging.Presented a poster paper at the WISH @ AMIA workshop and currently writing a full-length manuscript for JAMIA. | Jan 2025 – May 2026
Advisor: Dr. Michael Best |
| Georgia Institute of Technology
<i>Independent Research, D3 Lab</i> <ul style="list-style-type: none">Drafted and submitted the IRB protocol and designed a multi-method study investigating women's experiences with menopause in urban Pakistan.Conducted semi-structured interviews and walkthrough exercises with 13 women, a survey with 73 respondents, and interviews with 5 obstetrician-gynecologists.Synthesized findings using reflexive thematic analysis to characterize how space, place, and labor shape menopause experiences and identify implications for health technology design.Research published at ACM CHI 2026. | Jan 2025 – Dec 2025
Advisor: Dr. Naveena Karusala |
| Lahore University of Management Sciences
<i>Research Assistant, Interactive Media Lab</i> <ul style="list-style-type: none">Lead researcher for <u>Awaaz-e-Sehat</u>, a Gates Foundation-funded project selected through the Grand Challenges Catalyzing Equitable Artificial Intelligence initiative.Conducted ethnographic fieldwork and interviews in a gynecology department to understand clinical documentation workflows and co-designed the application's content and interface with clinicians.Developed and deployed an AI-based application using Bubble at a local hospital, integrating audio recording, speech-to-text transcription, electronic medical records, and automated identification of critical clinical risks.Published in ACM IMWUT; the project subsequently contributed to the formation of a healthcare startup in Pakistan. | Aug 2023 – Jul 2024
Advisor: Dr. Maryam Mustafa |

PRESENTATIONS

PEACH Nudge Dashboard: Eliciting Practitioner Needs for Targeted, Adaptive Health Messaging

Nov 2025

- *Workgroup on Interactive Systems in Healthcare (WISH) @ AMIA Workshop.*

INDUSTRY EXPERIENCE

Amazon Web Services (AWS)

May 2025 – Aug 2025

UX Research & Technical Writing Intern, Serverless Docs

Seattle, WA

- Conducted user research through surveys, interviews, and analytics to identify content gaps across the 900+ page AWS SAM Developer Guide, translating findings into content strategy and documentation recommendations.
- Designed and implemented a new content strategy, restructured key chapters, expanded examples, and improved onboarding, achieving 85% implementation during the internship.
- Produced measurable documentation improvements, including a 60% increase in positive feedback, an 11% reduction in negative feedback, and 81% higher engagement with restructured onboarding content.
- Recognized with a Think Big Award at the SPARK Intern Fair.

Taraki Technologies

Jul 2022 – Jul 2023

Product Associate, Product Experience Team

Lahore, Pakistan

- Directed MVP ideation and wireframing and tested prototypes with 15+ users to identify usability improvements for a job-search platform serving Pakistan's blue-collar workforce.
- Spearheaded 30+ A/B tests to evaluate interface components and optimize user flows based on behavioral patterns, contributing to a 46% improvement in conversion rates.

SELECTED APPLIED RESEARCH PROJECTS

Microsoft, Azure Database Platform Team

Aug 2025 – May 2026

Master's Capstone, UX Researcher

Atlanta, GA

- Conducted mixed-methods research with product designers through surveys, field studies, design workshops, and semi-structured interviews to identify barriers in AI-assisted documentation and requirements workflows.
- Designed a Copilot-powered concept for Microsoft Teams to capture, extract, and track design requirements from meetings and documentation, improving visibility into requirement changes across cross-functional teams.

Task Force for Global Health

Jan 2025 – May 2025

Industry Sponsored, Coalition for Global Hepatitis Elimination

Atlanta, GA

- Conducted mixed-methods global research for the COR-HEPB initiative, including content audits, surveys with 65 stakeholders, and task-based interviews across 28+ countries.
- Used findings to redesign the platform's information architecture and create a modular high-fidelity prototype.
- Evaluated the design through tree testing and moderated usability studies; the resulting platform was subsequently implemented and launched as a live resource for the global hepatitis B research community.

TEACHING EXPERIENCE

Georgia Institute of Technology

Jan 2025 – May 2025

Graduate Teaching Assistant, CS 3751: Introduction to UI Design

Atlanta, GA

- Instructed a studio section of 20+ students and mentored students throughout the user research and interaction design process.

Lahore University of Management Sciences

Sep 2022 – Dec 2022

Teaching Assistant, CS 466: Human-Computer Interaction

Lahore, Pakistan

- Supervised five student groups comprising 25 students on semester-long HCI projects.
- Conducted workshops on wireframing and interactive prototyping using Figma.

Lahore University of Management Sciences

Jan 2022 – May 2022

Head Teaching Assistant, CS 202: Data Structures

Lahore, Pakistan

- Helped introduce Python into CS 202 and designed programming assignments with automated test cases.
- Coordinated a team of 10 teaching assistants supporting a class of more than 200 students.

LEADERSHIP & SERVICE

MS-HCI Inclusion Committee

Sep 2024 – May 2026

Student Lead

Georgia Institute of Technology

- Represented students within the program and contributed to initiatives fostering an inclusive academic community.
- Co-created a professionalism pledge establishing shared community expectations within the MS-HCI program.

LUMS Women in Computing (LWiC)

Oct 2020 – Jun 2022

President

Lahore, Pakistan

- Led a student organization of 70+ members focused on increasing women's participation and inclusion in computing; led initiatives including a Python coding camp for high-school students.

Pakistan Women in Computing (PWIC)

Jun 2021 – Jun 2022

Student Ambassador

Lahore, Pakistan

- Selected for the Student Leaders Program to represent LUMS and develop an accessible local support network for students interested in computing.

RESEARCH METHODS & TECHNICAL SKILLS

Qualitative & Mixed Methods: Semi-structured interviews, literature reviews, survey design, usability testing, ethnography, contextual inquiry, think-aloud studies, heuristic evaluation, reflexive thematic analysis, tree testing, affinity mapping, user journey mapping, competitive analysis.

Design & Prototyping: Interaction design, wireframing, prototyping, user flows, storyboarding, mockups.

Programming: Python, C++ , JavaScript, HTML/CSS, SQL, PHP, MATLAB, LaTeX, Processing.

Tools & Analytics: Figma, Miro, Optimal Workshop, Mixpanel, Google Analytics, Bubble, Webflow, Git, Qualtrics.